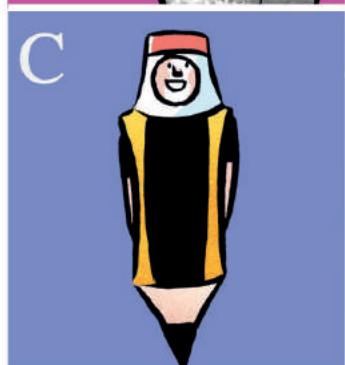




A
Periodic
Table that's
full of character
Meet the elements



April 20

With the support of:



This comic is adapted from the theatre show "The Dance of the Elements", an original idea by Adela Muñoz Páez, with texts by Pedro Coutteau, Sara María Pérez, Iván Espinosa, Ignacio Nieto, Ángela Romero, Alex Gómez-Reino, José M. García, María J. Vázquez, Pablo Ruíz, Francisco D. García, Alberto Benito, María Blasco, Adolfo Rosado, Melania Serrano, Palma García Celia López, Teresa Domínguez, Miguel Ruíz, Sergio Carrasco, Cristina Eslava, Jesús Aguilar, Marina García, Lucía Tocornal, Sara Ruíz, Juan Vidal and Juan F. Gracia.

Artwork and adaptation of script: Raquel Gu (@raquelberryfinn)

Coordination: Adela Muñoz Páez (@AdelaMunozUS) and Margarita Paneque Sosa

With the support of: M.^a Dolores Alcalá González, José Manuel Córdoba Gallego, Enrique Sánchez Marcos, Francisco Vega Narváez, University of Seville

English translation: Tim Gutteridge (@TimG_translator)

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METALS

- Alkali metal
- Alkaline earth
- Transition metal
- Lanthanide / Actinide
- Basic metal
- Semimetal

NON-METALS

- Non-metal
- Halogen
- Noble gas

1 H Hydrogen																	2 He Helium	
3 Li Lithium	4 Be Beryllium																	
11 Na Sodium	12 Mg Magnesium																	
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton	
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon	
55 Cs Cesium	56 Ba Barium																	
72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon				
87 Fr Francium	88 Ra Radium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson		

57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

The 150th anniversary of the creation of the Period Table, in 2019, provided a great opportunity to bring chemistry alive for young people, using fun activities to explain a science that has played a vital role in human progress..

We hope this initiative will inspire young people to enjoy and share the magic of chemistry.

Adela Muñoz Páez
Department of Inorganic Chemistry, University of Seville

Dmitri Mendeleev was born in Tobolsk, in Siberia, on 8 February 1834. He was the youngest of 17 children.

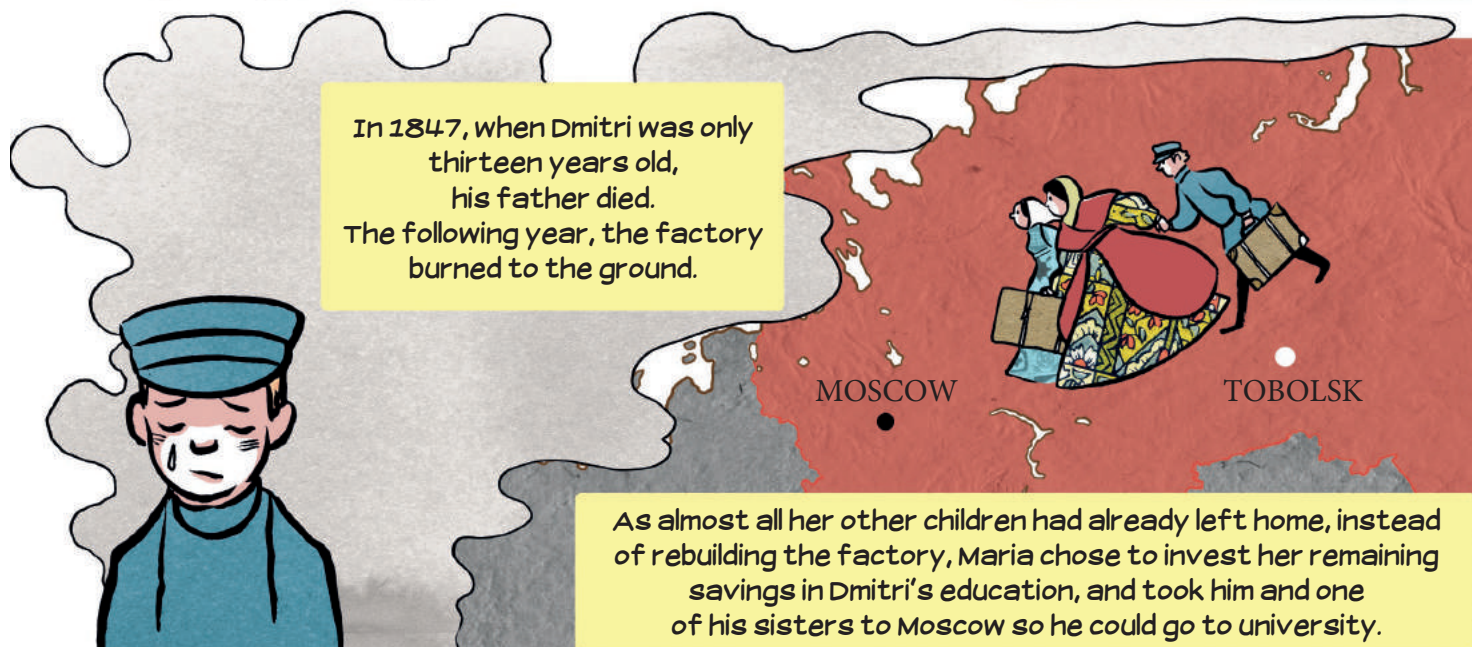


That same year, Dmitri's father, Ivan, who was the headmaster of the local school, went blind and lost his job. His mother, Maria, became the breadwinner of the family and assumed responsibility for running the glass factory, founded by Dmitri's grandfather.

"Mitienka", as Dmitri was known to his friends, stood out for his precocious talent for physics and mathematics.



In 1847, when Dmitri was only thirteen years old, his father died. The following year, the factory burned to the ground.



As almost all her other children had already left home, instead of rebuilding the factory, Maria chose to invest her remaining savings in Dmitri's education, and took him and one of his sisters to Moscow so he could go to university.

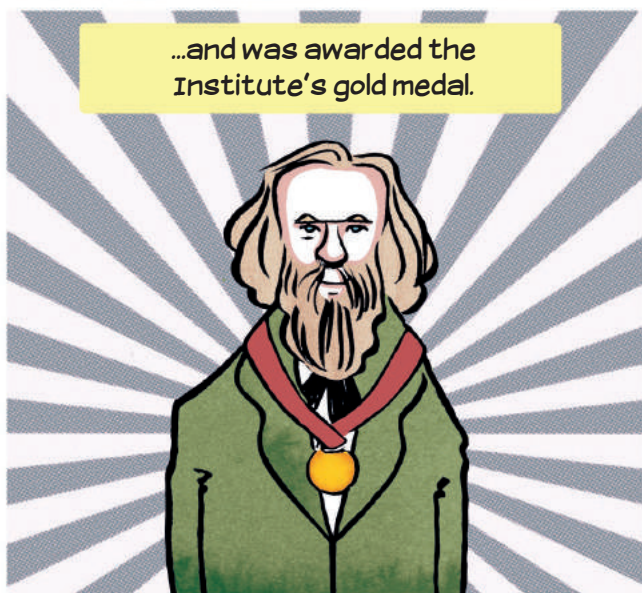
His Siberian origins meant he was not allowed to attend the University of Moscow or that of Saint Petersburg...



...however, he was admitted to the Central Pedagogical Institute of Saint Petersburg, where he completed his studies in physics and mathematics...



...and was awarded the Institute's gold medal.



At just 33 years of age, he was appointed Professor of General Chemistry at the University of Saint Petersburg, the most important chemistry professorship in Russia.



Mendeleev was an excellent teacher and, in order to help his students understand inorganic chemistry and to spare them the tedious task of memorizing endless lists of compounds and reactions, he identified a system of trends and patterns which would form the basis of a valuable learning resource: the Periodic Table.

		Ti=50	Zr=90	?=180.
		V=51	Nb=94	Ta=182.
		Cr=52	Mo=96	W=186.
		Mn=55	Rh=104.4	Pt=197.1.
		Fe=56	Ru=104.4	Ir=198.
		Ni=Co=59	Pd=106.6	Os=199.
		Cu=63.4	Ag=108	Hg=200.
H=1	Be=9.4	Mg=24	Zn=65.2	Cd=112
	B=11	Al=27.3	?=68	Ur=116
	C=12	Si=28	?=70	Sn=118
	N=14	P=31	As=75	Sb=122
	O=16	S=32	Se=79.4	Te=128?
	F=19	Cl=35.5	Br=80	I=127
Li=7	Na=23	K=39	Rb=85.4	Cs=133
		Ca=40	Sr=87.6	Ba=137
		?=45	Ce=92	
		?Er=56	La=94	
		?Yt=60	Di=95	
		?In=75.6	Th=118?	

Д. Менделѣевъ



The efforts of his mother Maria (widowed and nearly bankrupt) to support her son with all her strength to enable him to continue with his studies finally bore fruit.



Resist the temptation of illusion;
insist on work, not words;
search patiently for both
scientific and divine truth.



In 1869, some years after Maria's death, the Periodic Table was a reality: Dmitri organized the elements in rows and columns, making it easy to predict the chemical properties of each element based on those around it, establishing a template for their classification.

In expression of his eternal gratitude,
Dmitri dedicated his first book to his mother...



*This investigation is dedicated to the memory of a mother
by her youngest child. It was she who educated him, while
at the same time being responsible for managing a factory.
She taught him by example, corrected him with love and,
to enable him to dedicate himself to science, left Siberia
with him, spending what few resources and what little
strength remained.*

Hi! I'm Pedro but you can call me
Dmitri because I'm playing
Mendeleev in this celebration...

And I'm not alone.
I've brought some
of the elements
of the Table
along with me.

LET THE
PROCESSION
BEGIN!

...of the 150th
anniversary
of the
Periodic
Table.



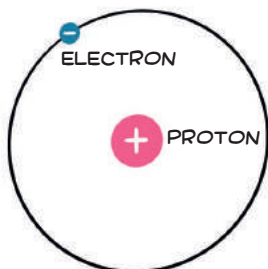
1
H
1.008

I'm hydrogen, the first element in the Periodic Table. I'm an invisible gas and I don't smell or taste of anything.

It doesn't sound like a whole load of fun but I'm one of the elements that make water – and you'll find me inside every living creature.



They call me simple!
Can you believe it?
Just because I only have
one proton and one
electron.



But I don't care, because it was thanks to my simplicity that I was the first element to form after the Big Bang.



What's more, I fuel the stars, powering the nuclear fusion that forms the other elements in the Periodic Table and giving off loads of energy. I'm the most abundant element in the universe!



I hope, from now on, every time you look up the sky and see the stars shining, you'll remember me.



3
Li
6.941

I'm lithium, a silvery-white metal, the lightest and softest there is.



I belong to the group of the alkali metals and, like all the members of that family, I'm very reactive, so you won't find me as a free element...

...but forming volcanic rocks and natural salts.



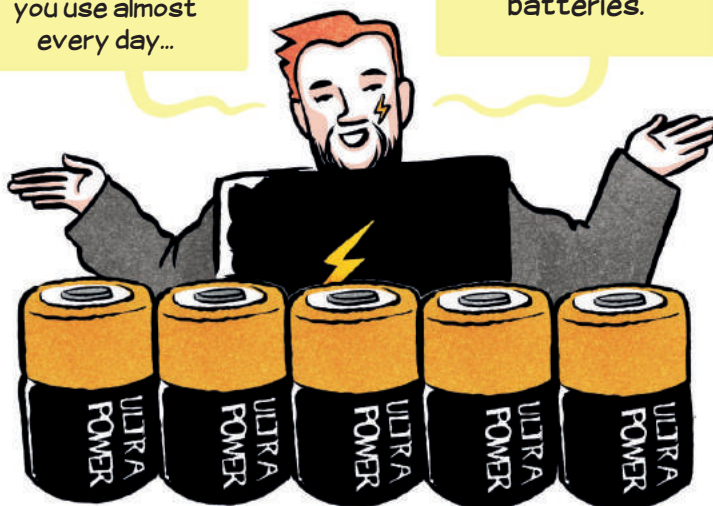
My name comes from the Greek word LITHOS, which means "stone".

And I was discovered by Swedish chemist Johan Arfwedson in 1817.

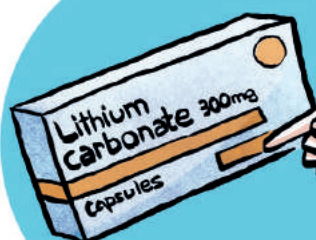


You can also find me in something you use almost every day...

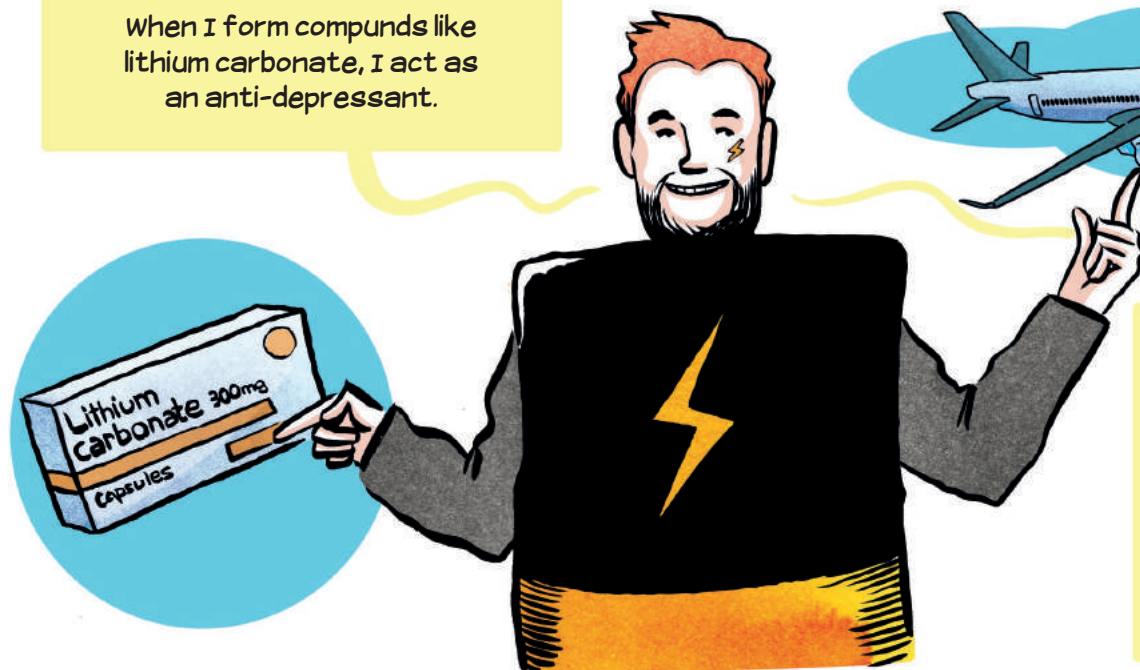
...long-life lithium batteries.



When I form compounds like lithium carbonate, I act as an anti-depressant.



I also form compounds with other metals such as aluminium, cadmium, copper or manganese, which are used in airplane construction.



5
B
10.811

I'm boron, one of those weird elements they call metalloids.



People have been using me for millennia. The Egyptians used me to mummify their dead; the Romans used me to make glass...



And I'm still around today. I make the green light in fireworks and I help make sure nuclear power stations don't explode.

But that's not all. Borosilicate glass is used to make glassware for laboratories. And plants love me because I help them stay in shape.



If you want to meet me, you'll need to look for my friends because you won't find me out and about on my own in nature.



If you're interested, just ask for Borax.

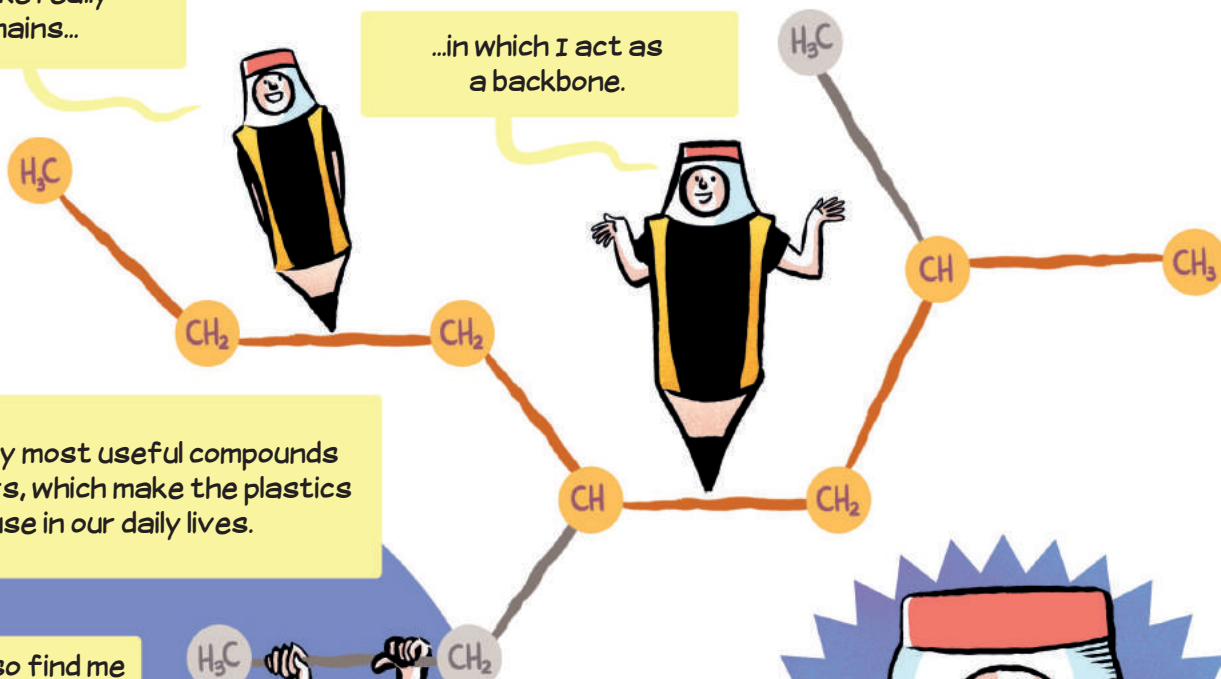
6
C
12.011

I'm carbon and without me life wouldn't even exist. I can be found in all living beings because I'm used to make proteins, vitamins, carbohydrates and nucleic acids (DNA, RNA...).



You'll also find me in pencils, in just about anything you eat, and in the medicines you take when you're sick. And I make beautiful structures like diamonds and useful ones like coal.

I can make really long chains...



Some of my most useful compounds are polymers, which make the plastics we use in our daily lives.

And you'll also find me in petrol and packaging, in cars and planes, in clothes and telephones...



So you tell me if I'm useful or not!



7
N
14.007

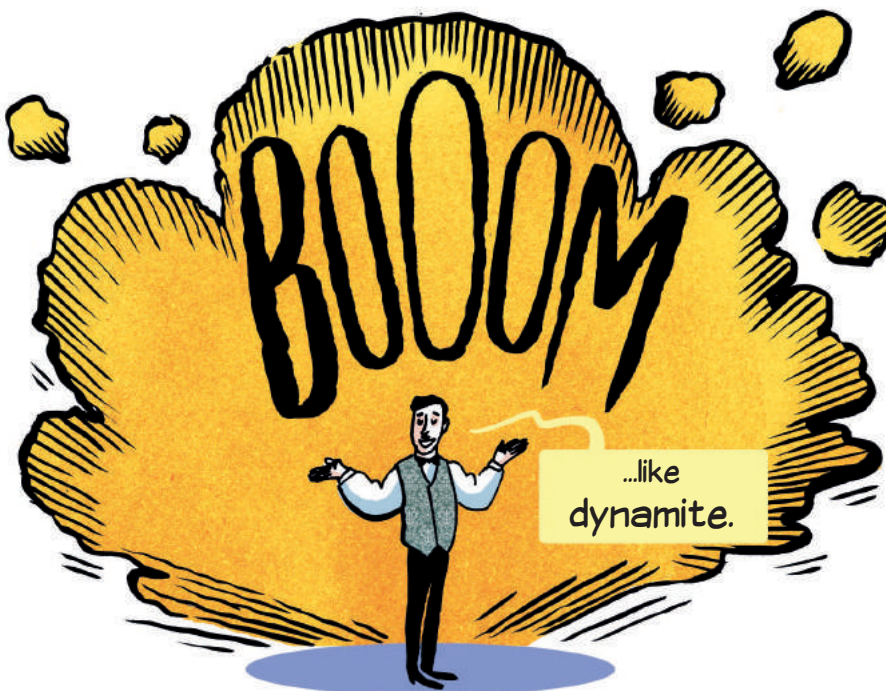
I'm nitrogen, a gaseous element with the atomic symbol N.



I like to hang out with my friends in the earth's atmosphere, creating the colours of the rainbow and beautiful sunsets.

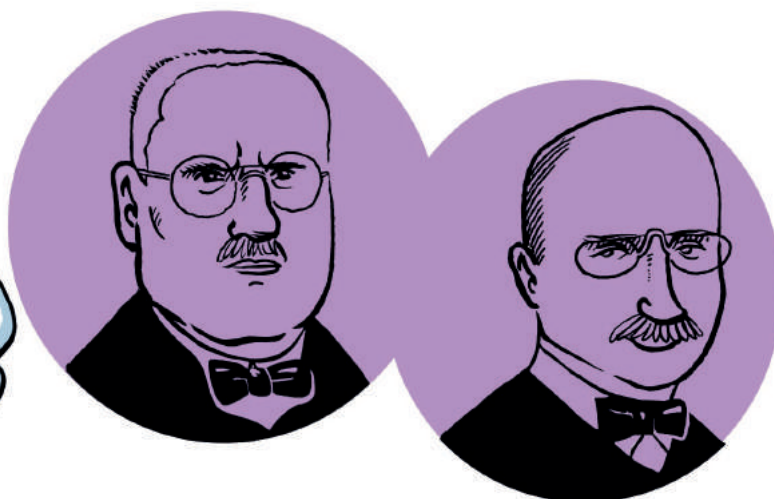


I bet you think that sounds lovely! And it's true that I'm essential for life. But I've got a dangerous side too. I pop up in lots of toxic and explosive compounds...



...like dynamite.

I have applications in fields as diverse as agriculture and gastronomy, but my cleverest trick of all is bonding with hydrogen to make ammonia, the most important fertilizer there is.



And I'm wearing my smartest clothes in honour of the two scientists who patented this process: Carl Bosch and Fritz Haber.

9
F
18.998

I'm fluorine. My atomic number is 9 and I'm the first of the halogens.



I'm the most electronegative element in the table and I can form compounds with just about any other element, including some of the noble gases!



It took people centuries of trying but Frenchman, Henri Moisan, finally discovered me in 1886 when he managed to isolate me in a pure state.



I'm so toxic that I accidentally killed several of the scientists who worked on me. Whoops!



You might wonder what use I am if I'm so dangerous... But you couldn't live without me!

I'm essential for making bones strong - and for hardening the enamel on your teeth!



And without me it would be impossible to obtain aluminium or the uranium used to generate nuclear power.



15
P
30.974

I'm phosphorus and I've just arrived from the cells, transporting packets of energy.



I was born in 1669, when my father, Hennig Brand, was searching for the philosopher's stone...



...and he decided to take some of his urine, mix it with sand...

...and heat it all up.

The result was a white solid which glowed in the dark, so Hennig called me "phosphorus", from the Greek PHOS (light) and PHORUS (carrier).



I'm used in the manufacture of matches. But before I can do that I have to put in some hard work... sunbathing with my friend calcium.

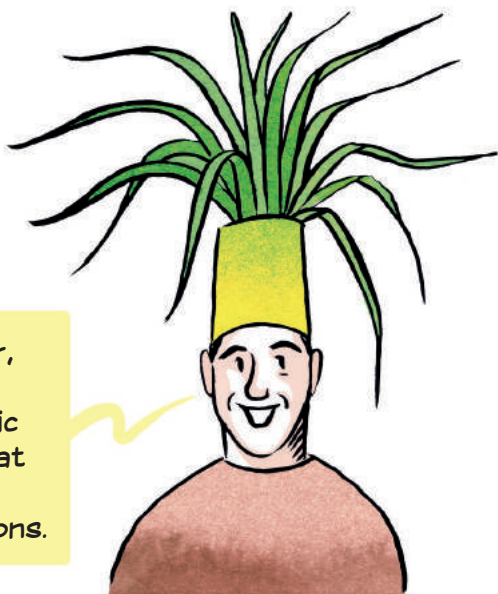
Hey, calcium! Come and catch some rays with me. I'm getting nice and red so I can be used in matches; if I don't get the sun I go so pale that I become poisonous.

What's that, calcium?

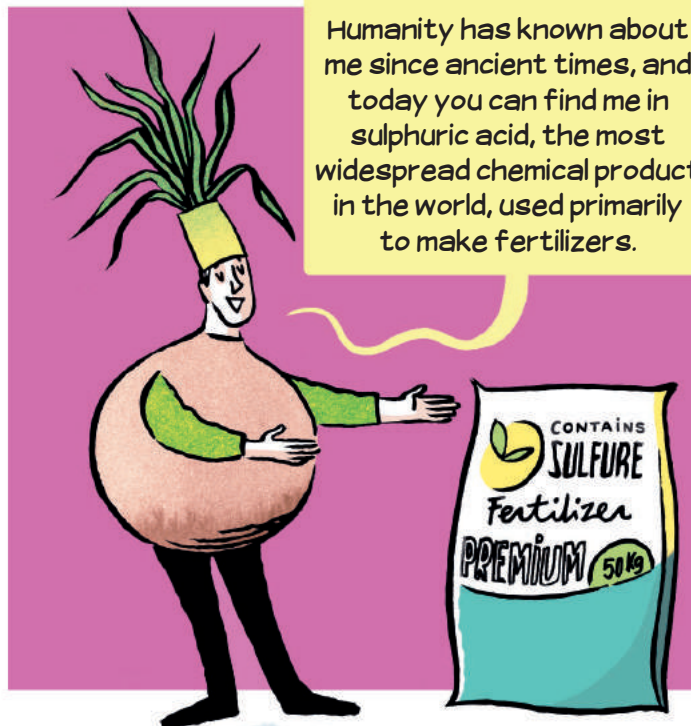
A cellular emergency? I'm on my way. They need some phosphorylation!



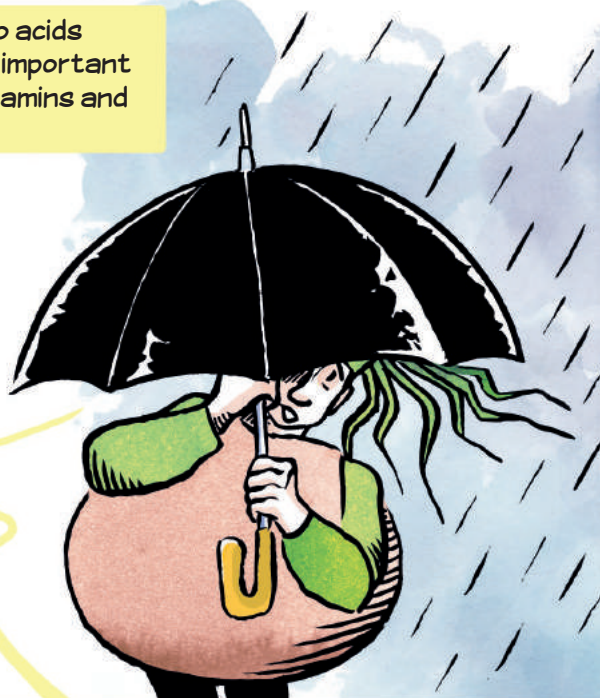
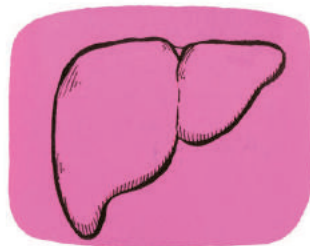
I'm sulphur,
a yellow
non-metallic
element that
occurs in
volcanic regions.



Humanity has known about
me since ancient times, and
today you can find me in
sulphuric acid, the most
widespread chemical product
in the world, used primarily
to make fertilizers.



I form a part of two fundamental amino acids
- cysteine and methionine - which are very important
in proteins. You can also find me in some vitamins and
I help the liver to secrete bile.



I'm to blame for acid rain...

...and for making you cry when you chop onions.
That's because sulphur compounds are released
when you cut through the onion cells. And when the
compounds come into contact with the moisture
in your eyes, sulphuric acid is formed,
causing your eyes to sting.



I'm potassium, an essential element for life, and I'm never alone - I'm always making compounds.

You'll find me in fertilizers, fireworks and gunpowder.



If you pour water on me, you'll get a really explosive reaction:
I GO BANANAS!!!

I've even been used to execute people by injecting them with potassium chloride.

But I'm not all bad. I stabilize DNA and RNA in living creatures.
And I help growth and respiration in plants.

Hey, just like me!

I also pop up in sodium-potassium pumps, a mechanism that regulates osmotic equilibrium in cells.

And as you've probably guessed from my outfit, I'm in bananas!

I'm in lots of other foods too, and it's really important to make sure there's enough of me in your diet.



20
Ca
40.078

I'm calcium,
a vital component
of your bones.

I'm pretty
sociable.



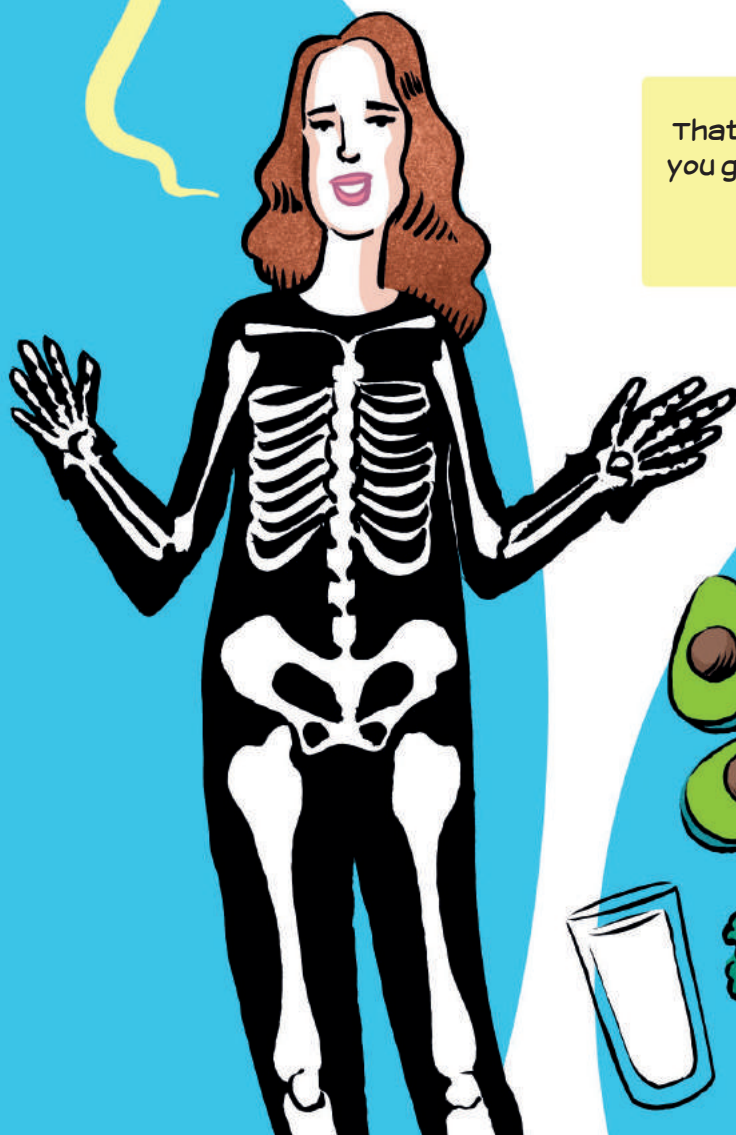
I like getting on with other elements.
When I link up with sulphates I form natural chalk,
which is used to make cement or to put a plaster
cast on broken arms or legs. I'm also in calcium
carbonate, which is the main component
of seashells...



But my most important relationship
is the one I have with phosphate...
to make your skeletons!

And calcium phosphate makes up a lot of
your dental enamel, too.

That's why it's really important
you get plenty of me in your diet.
I'm more useful than
you might think!



26
Fe
55.933

I'm iron, an element with very interesting properties...



And I can act as a matrix for other elements that give me different properties. For example, steel is made by combining me with a little bit of carbon.



I'm iron, the fierce FERROUS wolf, and the rocks are the carbon atoms.

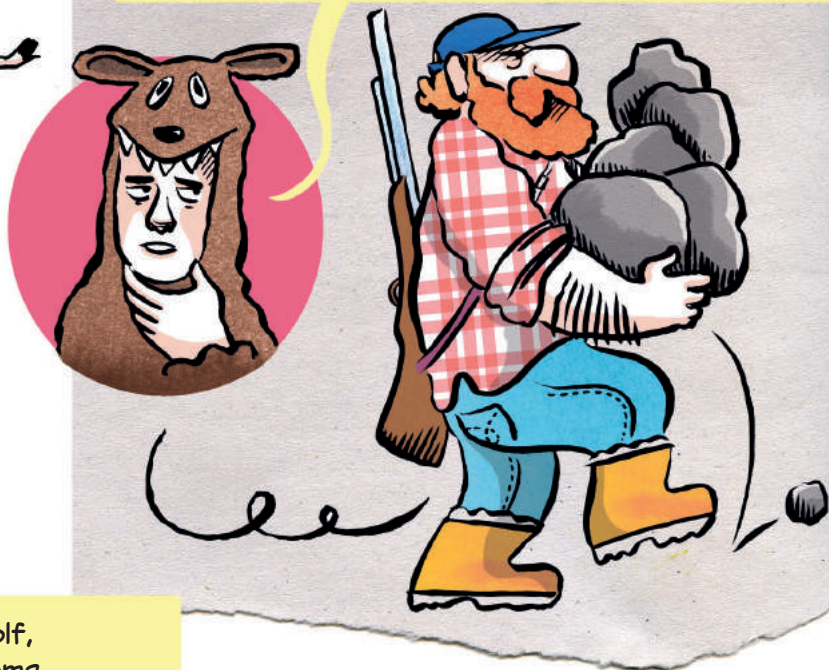


I'm the fourth most abundant metal on earth. I'm very malleable, which means I can change shape if I'm worked in a forge.

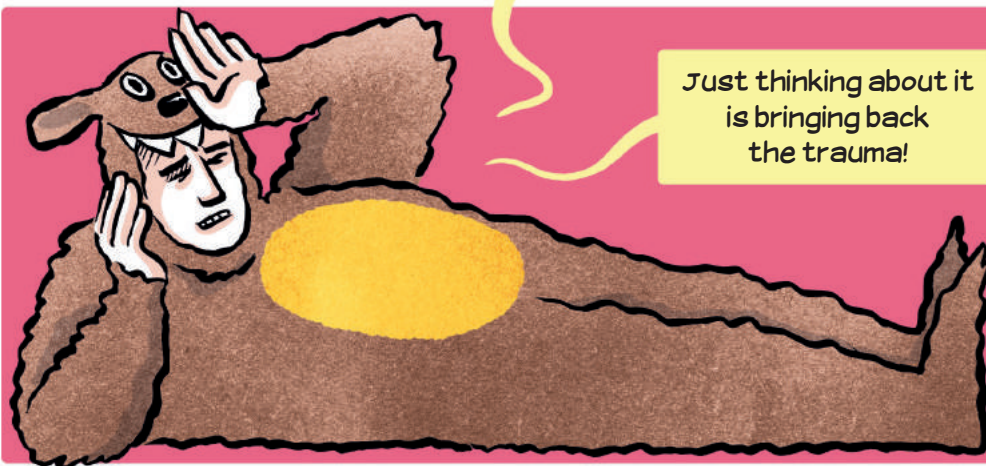
I have magnetic properties.



Do you remember the story of Little Red Riding Hood? At the end, the hunter filled my stomach with rocks to make me sink.



Just thinking about it is bringing back the trauma!



31

Ga

69.723



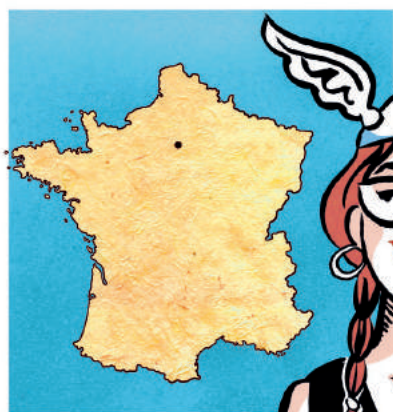
I'm gallium,
a shiny silvery metal
that is usually solid...

...although in summer, when it
gets hot, I start to melt!
The heat of your hand is
enough to turn me to liquid.



I was discovered in France, which used to be called Gaul,
so that's where my name comes from. And you could hardly
sound more French than the man who discovered me...

Paul Émile Lecoq de Boisbaudran.



I don't exist freely in nature but you can
obtain me when you make my friend, aluminium.
If it wasn't for me, you wouldn't be able to
watch TV or use those devices you're so
dependent on. That's right: mobile phones!!!



One last piece
of advice.
Whatever you do,
don't shut me
in a tube.
I suffer from
claustrophobia,
the fear makes me
expand and...
BOOOM!



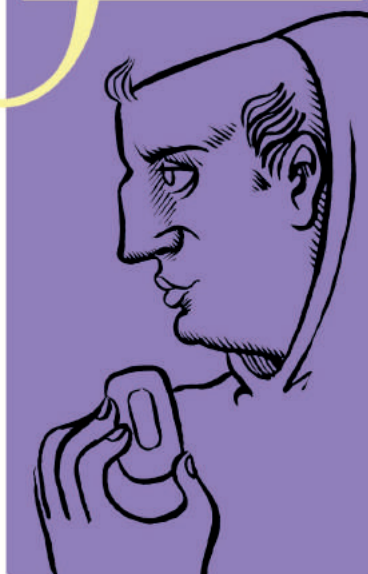
As

74.922

I'm arsenic but all my friends call me As...



I was discovered by Saint Albert the Great when he heated soap with arsenic mineral.



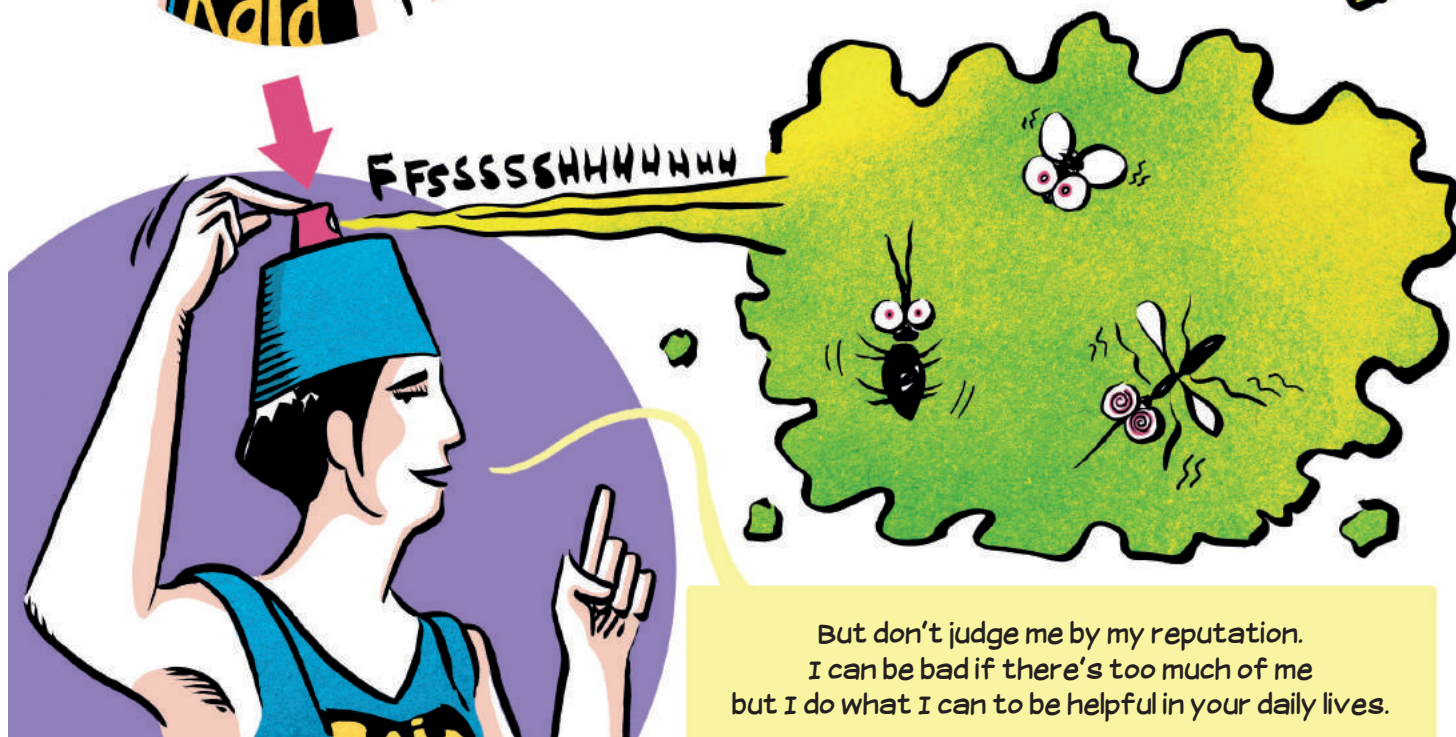
I crop up as a poison in lots of books and films but I have plenty of other uses too.



I strengthen copper and lead alloys, and improve the properties of glass. I'm really useful in the manufacture of semi-conductors, and I stop wood from going rotten. What's more, I'm used to treat cancer, leukaemia and psoriasis.



I'm also in herbicides, pesticides and insecticides, although you won't find me in domestic products because I'm just a teensy bit toxic...



But don't judge me by my reputation. I can be bad if there's too much of me but I do what I can to be helpful in your daily lives.

36

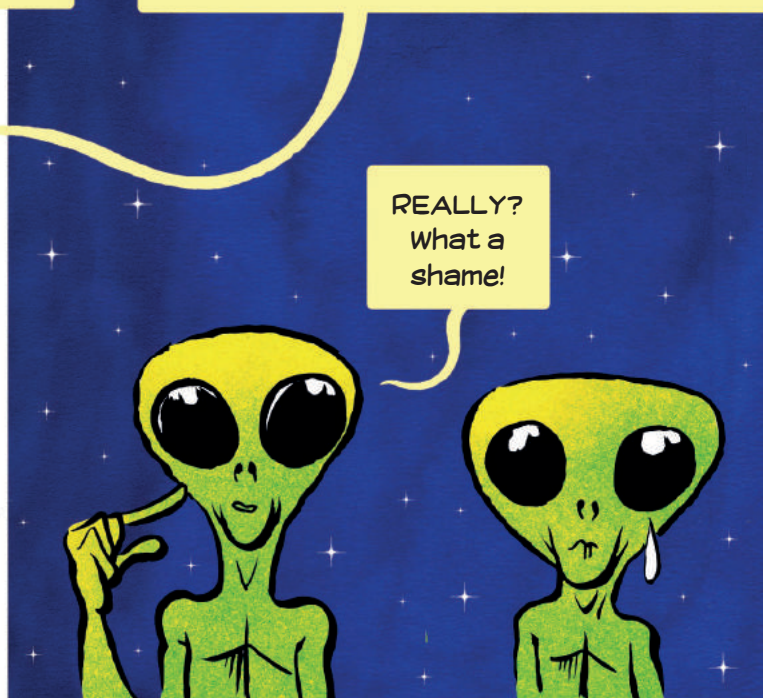
Kr

83.798

I'm the Hidden One, although everyone knows me as krypton, which comes from the Greek KRYPTOS, meaning "cryptic"...



You can find me in volcanic gases, thermal waters and also in some minerals and meteorites, but I don't have anything to do with aliens!



REALLY?
What a
shame!

I'm less dense than air, so – just like helium – if you inhale me your voice will go all squeaky. But be careful, because if you breathe in too much of me you can die from asphyxiation.



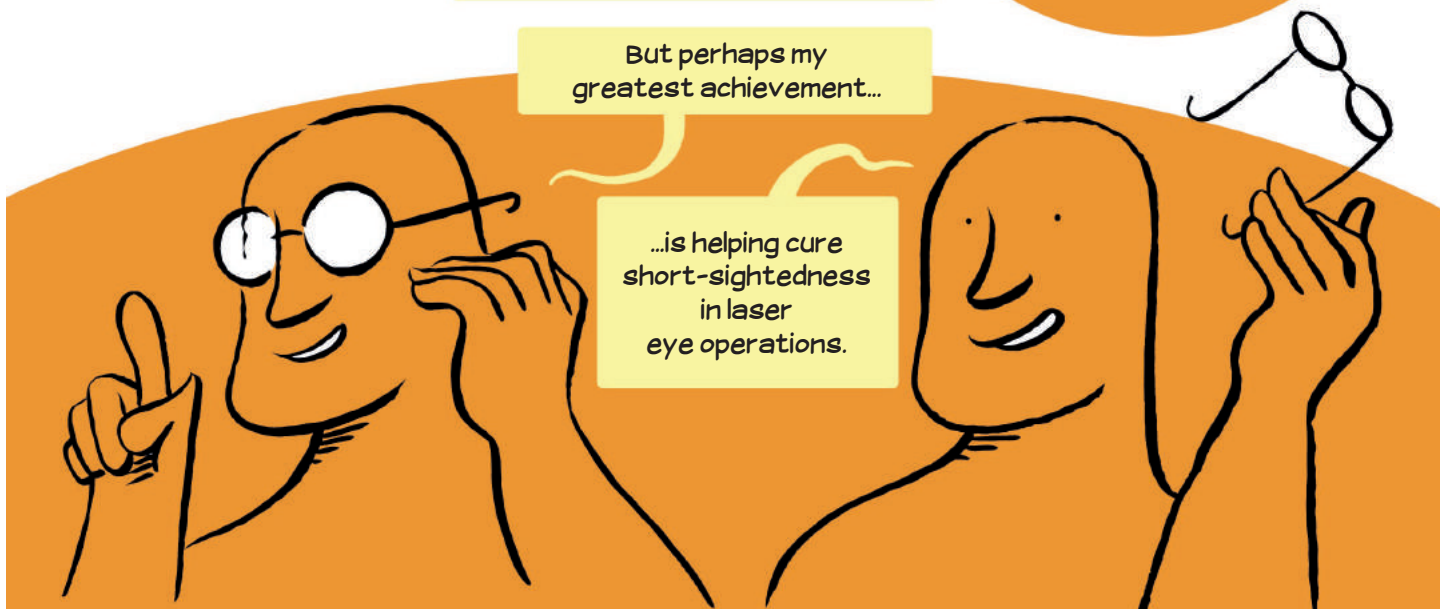
You might be surprised to know that I pop up quite a lot in your lives: I help make the flashes used for taking high-speed photographs...

...and I'm in the lights on airport runways, because my light is visible from a long way off, even when it's foggy.



But perhaps my greatest achievement...

...is helping cure short-sightedness in laser eye operations.



I'm here to bring you all a little glamour: I'm silver and my symbol is Ag, which comes from the Latin ARGENTUM and means "shining". Because I'm worth it!!!

47

Ag

107.868

Back in the old days, they attributed divine properties to people who had contact with me... and who could blame them!

I was also an essential component of coins..

Today, I'm used to make jewellery, because I don't rust.

But my virtues don't end there. I'm very good at conducting electricity, which is why you'll find me in all those gadgets that everyone loves so much...

I'd love to stay and tell you more about myself but I'm afraid I've got to rush. As you can see, I'm pretty popular.

Ciao, darlings!

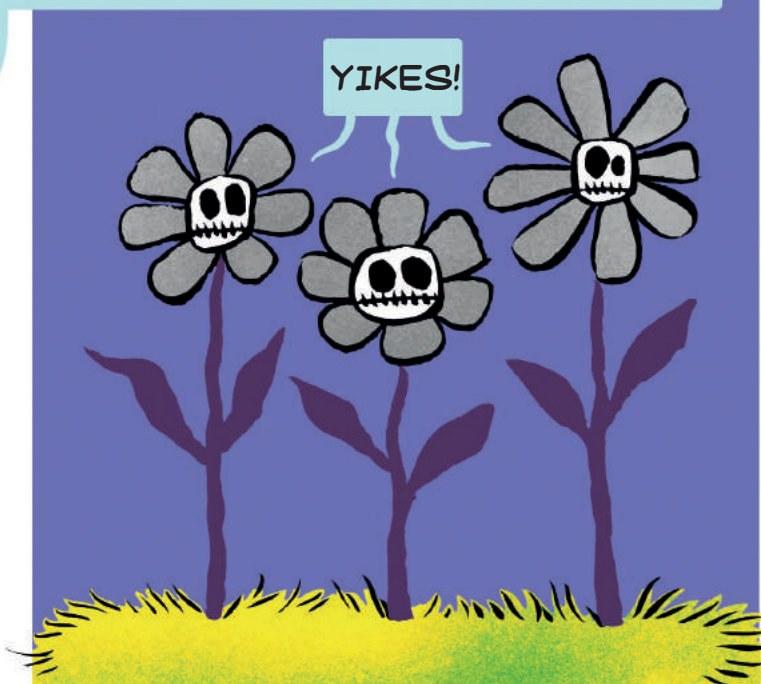


Cd

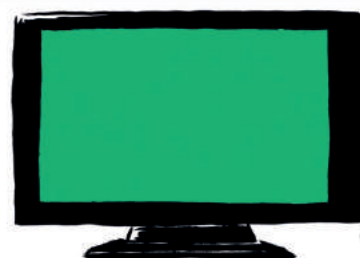
112.411



I'm cadmium, one of the most dangerous toxic metals of all, because plants absorb me from the soil.



You can find me in all sorts of daily objects, like the disposable batteries in toys and remote controls, and also in TVs, where I help produce blue and green light... Oh, what lovely colours!



I'm dressed as an artist because I'm in lots of pigments, such as yellow, where I appear as cadmium sulphide which is...

...where my discoverer - a German chemist called Friedrich Strohmeyer - found me. I also crop up in bright red and orange pigments.

And because those pigments are heatproof, I'm used on crockery and tableware. Be careful if it smashes though or I might end up poisoning you!



I'm indium and my name comes from the indigo colour of my bright blue spectral line... but I also like dressing up as an American Indian!



I was discovered by two German chemists, Ferdinand Reich and Hieronymus Theodor Richter but...



...the pair fell out when Richter tried to take sole credit for their achievement.



I'm silver-coloured and I'm one of the softest elements in the Periodic Table, and I'm great at forming alloys. I've been used for all sorts of things. During WWII, I was used to coat bearings in aircraft engines but nowadays you're most likely to find me in LCD screens for TVs and computers.

If you bend me,
I let out a high-pitched scream!

I'm not particularly abundant and I hardly ever appear on my own, so I have to be extracted from other substances when they're smelted down to produce other metal ores such as zinc.



57

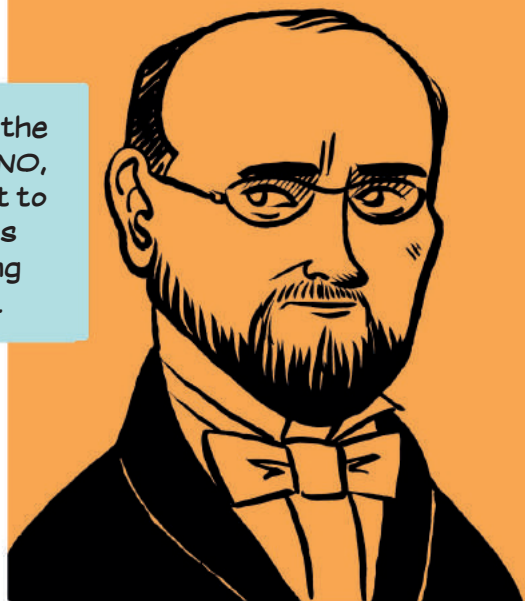
La

138.906

I'm lanthanum, and I give my name to a whole family of elements: the lanthanides.

I'm the first of the rare earth elements, and one of the most reactive metals in the group.

My name comes from the Greek word LANTHANO, which means "difficult to find", because I was finally spotted hiding in a cerium mineral.



I was discovered by Swedish chemist Carl Mosander in 1839, although it wasn't until 1923 that I was obtained in my pure form.

I'm an important element in glass manufacture because I have a high refractive index, which means light slows down and bends when it passes through me.



And that's why I'm also used in the movie industry, in lighting, and in the manufacture of lenses and components for video cameras.



71

Lu

174.967

I'm lutetium and I'm one of the lanthanides.

I was discovered in 1907 by a Frenchman, Georges Urbain, and an Austrian, Carl Auer von Welsbach.



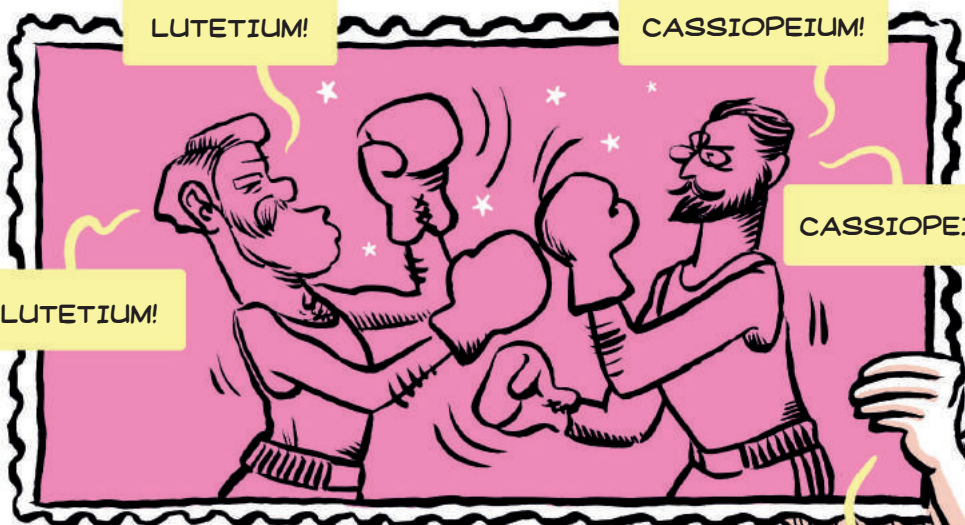
Urbain and von Welsbach argued about what to call me. The Austrian wanted to call me cassiopeium, in honour of the constellation Cassiopeia, and the Frenchman wanted to call me lutetium, from Lutetia, the Latin name for Paris...

LUTETIUM!

CASSIOPEIUM!

LUTETIUM!

CASSIOPEIUM!



It's clear who won...

I'm used as a catalyst in chemical reactions such as hydrogenation, cracking, polymerization and alkylation. But my favourite job of all is when one of my isotopes is used to treat tumours.

People don't usually pay me much attention but, as you can see, I'm pretty useful and I have a rather interesting history.



Ta

180.948

My name is tantalum, and I'm a hard, malleable, corrosion-resistant metal.

I was discovered by Swedish chemist Anders Gustaf Ekeberg in 1802, but until 1844 everyone thought I was the same as niobium.

Without me, you wouldn't have computers, mobile phones or a whole load of other electronic gadgets.



Because I'm obtained from tantalite, a mineral found in coltan, which is mined...



...in countries at war, exploiting thousands of people, there's a lot of controversy surrounding my use and the need to recycle me.



I was named after Tantalus, a figure from Greek mythology, condemned to suffer eternal hunger in Tartarus, tied to a huge tree for offending his father, Zeus, killing his own son and deceiving other gods...

Now if you'll excuse me, I'd better get back to Tartarus before Zeus realizes I've nipped out.

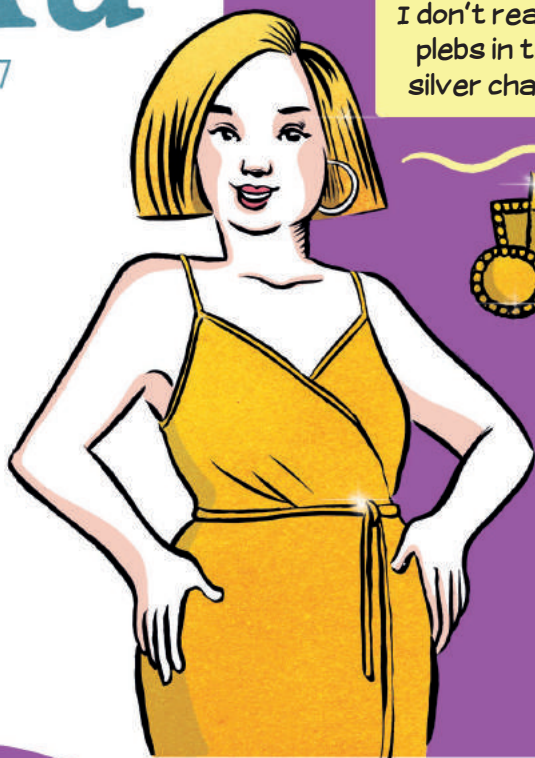


Au

196.967

I'm gold and my chemical symbol is Au, which comes from the Latin AURUM, meaning "radiant dawn". I've been a very important metal in many civilizations - a symbol of royalty, purity and bravery.

I don't react with other elements. I don't like mixing with all those plebs in the Periodic Table. And certainly not with that ghastly silver character, who's always behaving as if she was my equal!



I can form fine sheets or be stretched into thin threads so I can be used to create stunning jewellery... but I'm also a dab hand at electronics. Your mobile phones, for example, wouldn't work without me...

Because I'm so valuable, back in the old days alchemists kept trying to make me out of elements such as lead...

...but it didn't work because I'm one of a kind. I hope you all consider yourselves lucky I made the time to come along tonight!



Hg

200.590

I'm mercury and you're probably wondering why I'm dressed as the Mad Hatter from "Alice in Wonderland". That's because hatmakers were driven crazy by an element named after the messenger of the gods...



In the old days I was called quicksilver because I looked like liquid silver - HYDRARGYUM in Latin - which is why my symbol is Hg.



I enter the body through the nose, the mouth or the skin and, before I started turning hatters mad, I attacked alchemists like Isaac Newton, tsars like Ivan the Terrible, and even Napoleon...



...who was poisoned when the dose of his medicine for stomach pain was gradually increased.



I'm so toxic they've banned me! But before that happened, I could be found in thermometers - where I helped tell doctors if their patients were sick - and in the fillings in people's teeth.



Hatmakers used me to make felt for their creations. But they were in poorly ventilated workshops and the mercury vapour gradually poisoned them and caused them to lose their minds... Which is where the phrase "as mad as a hatter" comes from!



Fr

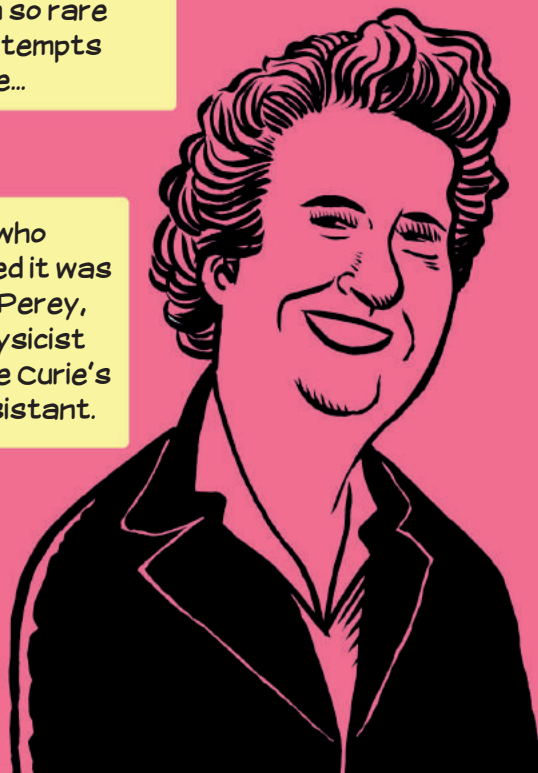
223.020



I'm francium and I'm so rare that it took four attempts to discover me...

The one who finally managed it was Marguerite Perey, a French physicist who was Marie Curie's personal assistant.

And she gave me this name in honour of my country of birth!



People say I can be a bit heavy - but I think they're just referring to my chemical properties! Almost all my salts are soluble in water, which is just as well at the height of summer. I have very high radioactivity and my half-life is just... 22 minutes!



Researchers have used me as a guinea pig in the field of spectroscopy, but outside the laboratory I'm not a great deal of use... to be frank.

So if you're looking for me, you'll know where to find me!



"Au revoir my little chums!"



92

U

238.029



I'm uranium - U to my friends -
and I belong to the actinide series.

My discoverer,
Martin Heinrich
Klaproth, named me
after Uranus...
I just hope he asked
for your permission!
I occur naturally
when supernovas
explode.

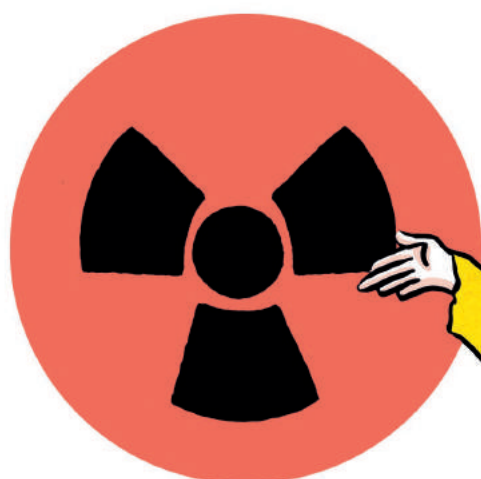


I have the highest atomic mass
of any naturally occurring element.
Because of my radioactive
properties, I'm used as fuel in
nuclear power plants.

And because my atoms
decay so slowly, I can be
used to calculate the age of
the planet earth,
this home of ours that
we need to look after.



Although I'm best known for my radioactivity, I'm even more toxic
as a heavy metal!



95

Am

243.061

I'm americium and I was discovered by mistake after a nuclear test in 1944, in Chicago, USA - hence my name. But I was kept secret until November 1945, just in case the Soviets found out.



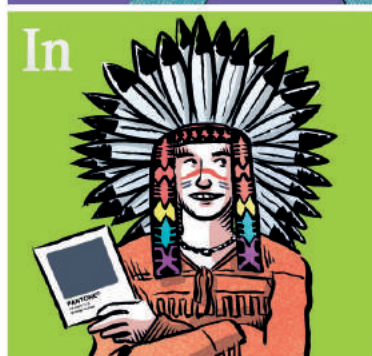
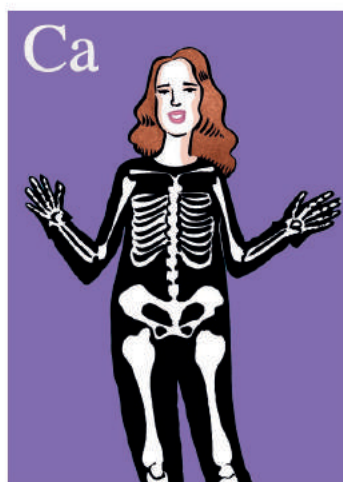
I belong to a very radioactive family, the actinides, and that means I'm dangerous to be around. If you swallow me by mistake, I deposit myself in your bones and start to emit particles at high speed, causing a mutation that leads to bone cancer.



Despite that, I was used in smoke detectors for a long time, because I react so easily with CO₂. I'm also the only synthetic element that can be purchased commercially!

So there you have it...
the
Americium
Dream!





GOBIERNO
DE ESPAÑA

MINISTERIO
DE CIENCIA
E INNOVACIÓN



FUNDACIÓN ESPAÑOLA
PARA LA CIENCIA
Y LA TECNOLOGÍA

Andrés
20